

6 (a) (i) From Fig. 6.2, the current at $t = 0.070$ s is 1.500 A.

$$n = \frac{3000}{0.15} = 20000$$

M1

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 20000 \times 1.500$$

M1

$$= 3.77 \times 10^{-2} \text{ T}$$

A0

(ii) Using points (0.070, 1.500) and (3.80, 0.550) on the line in Fig. 5.2,

$$\frac{dI}{dt} = \frac{0.550 - 1.500}{0.380 - 0.070} = -3.0645 = -3.06$$

C1

According to Faraday's law,

$$\text{e.m.f.} = -\frac{d\Phi}{dt} = -\frac{d}{dt}(N_c B_s A_s)$$

$$= -N_c A_s \frac{d}{dt}(\mu_0 n I)$$

$$= -N_c A_s \mu_0 n \frac{dI}{dt}$$

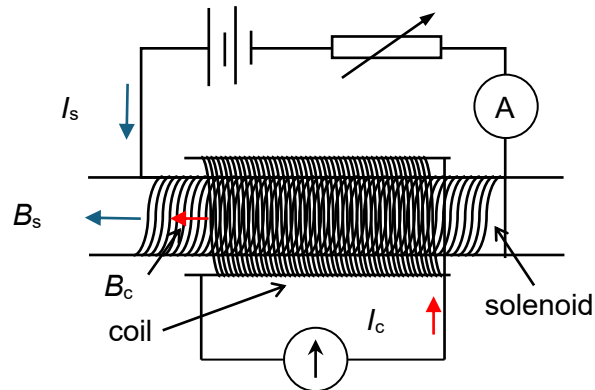
$$= -1500 \times 2.5 \times 10^{-4} \times 4\pi \times 10^{-7} \times 20000 \times (-3.0645)$$

C1

$$= 2.8882 \times 10^{-2} = 2.89 \times 10^{-2} \text{ V}$$

A1

(iii)



The current I_s in the solenoid flows from the positive terminal of the battery.
By right-hand-grip rule, the magnetic flux density B_s produced points to the left.

B1

Since I_s decreases, B_s also decreases so by Lenz's law, the induced magnetic flux density B_c in the coil must point to the left.

B1

By right-hand-grip rule, the induced current I_c in the coil flows from left to right through the galvanometer.

B1